

These notes derive the stability analysis of the replica-symmetric saddle point of the Sherrington–Kirkpatrick (SK) model (Sherrington and Kirkpatrick, 1975): the structure of the fluctuation Hessian and its exact diagonalization by the representation theory of the permutation group (longitudinal, anomalous and replicon sectors); the replica-symmetric free energy and its negative zero-temperature entropy $S(0) = -1/2\pi$; the de Almeida–Thouless (AT) instability (de Almeida and Thouless, 1978); the meaning of saddle points in the replica limit; the Parisi construction of replica-symmetry breaking (1RSB, k RSB, full RSB); the overlap distribution and ultrametricity; the physical meaning of the three sectors; and the fate of the replicon on the Nishimori line, where gauge identities enforce $\chi_{\text{SG}} = \chi_{\text{F}}$. Standard references are Mézard et al. (1987), Fischer and Hertz (1991) and Nishimori (2001). Solutions to the exercises are collected at the end.

1 The SK model and the replica method

The SK model consists of N Ising spins $s_i = \pm 1$ with Hamiltonian

$$H = - \sum_{i < j} J_{ij} s_i s_j - h \sum_i s_i, \quad J_{ij} \sim \mathcal{N}(0, \frac{1}{N}) \text{ i.i.d.} \quad (1.1)$$

The quenched free energy requires $\mathbb{E}[\ln Z]$, which the replica method obtains from integer moments,

$$\mathbb{E}[\ln Z] = \lim_{n \rightarrow 0} \frac{1}{n} \ln \mathbb{E}[Z^n]. \quad (1.2)$$

For integer n , replicate the spins, $s_i \rightarrow s_i^a$, $a = 1, \dots, n$, and average over the Gaussian couplings using $\mathbb{E}[e^{xJ}] = e^{x^2/2N}$ for each bond:

$$\mathbb{E}[Z^n] = \text{Tr} \exp \left[\frac{\beta^2}{2N} \sum_{i < j} \left(\sum_a s_i^a s_j^a \right)^2 + \beta h \sum_{i,a} s_i^a \right]. \quad (1.3)$$

Using $(\sum_a s_i^a s_j^a)^2 = n + 2 \sum_{a < b} (s_i^a s_i^b)(s_j^a s_j^b)$ and decoupling the sites with a Hubbard–Stratonovich field Q_{ab} for each replica pair $a < b$,

$$\mathbb{E}[Z^n] = e^{N\beta^2 n/4} \int \prod_{a < b} dQ_{ab} \exp \left[- \frac{N\beta^2}{2} \sum_{a < b} Q_{ab}^2 + N \ln \text{Tr}_{\{s^a\}} e^{L(Q)} \right], \quad (1.4)$$

$$L(Q) = \beta^2 \sum_{a < b} Q_{ab} s^a s^b + \beta h \sum_a s^a, \quad (1.5)$$

which is now a single-site problem. The order parameter is the overlap matrix Q_{ab} , a symmetric matrix with no diagonal. Its independent components are the $\binom{R}{2} = R(R-1)/2$ pairs $a < b$, where from here on we write R for the number of replicas and reserve $n \rightarrow 0$ for the replica limit. This counting is the origin of every dimension formula below.

2 The replica-symmetric saddle point

The saddle point of Eq. (1.4) is

$$Q_{ab} = \langle s^a s^b \rangle_L, \quad (2.1)$$

where $\langle \cdot \rangle_L$ is the single-site average with weight e^L . The replica-symmetric (RS) ansatz is $Q_{ab} = q$ for all $a < b$. Then $\sum_{a < b} s^a s^b = \frac{1}{2}[(\sum_a s^a)^2 - R]$, and a second Hubbard–Stratonovich transformation linearizes the square:

$$\text{Tr } e^L = e^{-\beta^2 q R/2} \int \mathcal{D}z [2 \cosh \Xi(z)]^R, \quad \Xi(z) = \beta(\sqrt{q}z + h), \quad (2.2)$$

with $\mathcal{D}z = \frac{dz}{\sqrt{2\pi}} e^{-z^2/2}$. In the replica limit $R \rightarrow 0$ the measure $[2 \cosh \Xi]^R \rightarrow 1$, and the single-site correlators needed below reduce to moments of $\tanh$:

$$q = \int \mathcal{D}z \tanh^2 \Xi(z), \quad r \equiv \langle s^a s^b s^c s^d \rangle = \int \mathcal{D}z \tanh^4 \Xi(z) \quad (a, b, c, d \text{ distinct}). \quad (2.3)$$

The first relation is the RS self-consistency equation. Note also $\langle s^a s^b s^a s^c \rangle = \langle s^b s^c \rangle = q$ because $(s^a)^2 = 1$: a four-spin correlator with one repeated replica index reduces to a two-spin one. This identity determines the whole structure of the Hessian.

Laplace evaluation and the RS free energy

In this subsection we restore the coupling scale: J_{ij} has variance J^2/N , which replaces $\beta \rightarrow \beta J$ in every formula above (the Hessian sections use $J = 1$ again). The Q -integral in Eq. (1.4) is of Laplace type, $\int dQ e^{Ng(Q)}$; for large N ,

$$\int dQ e^{Ng(Q)} = e^{Ng(Q^*)} \sqrt{\frac{2\pi}{N|g''(Q^*)|}} [1 + O(1/N)], \quad (2.4)$$

so the free-energy density is the saddle value $g(Q^*)$ up to $O(\ln N/N)$ corrections. (Strictly, one should take $N \rightarrow \infty$ before $n \rightarrow 0$; as usual we exchange the limits.)

Insert the RS point $Q_{ab} = q$ at $h = 0$, using $\sum_{a < b} q^2 = \frac{n(n-1)}{2} q^2$ and the single-site trace (2.2):

$$\frac{1}{N} \ln \mathbb{E}[Z^n] = \frac{(\beta J)^2 n}{4} - \frac{(\beta J)^2 n(n-1)}{4} q^2 - \frac{(\beta J)^2 q n}{2} + \ln \int \mathcal{D}z [2 \cosh \Xi(z)]^n, \quad \Xi = \beta J \sqrt{q} z. \quad (2.5)$$

Expand $[2 \cosh \Xi]^n = e^{n \ln 2 \cosh \Xi} = 1 + n \ln 2 \cosh \Xi + O(n^2)$ and $\ln(1 + nx) = nx + O(n^2)$, divide by n and take $n \rightarrow 0$:

$$-\beta f_{\text{RS}} = \frac{(\beta J)^2}{4} (1 - 2q + q^2) + \int \mathcal{D}z \ln 2 \cosh \Xi(z), \quad (2.6)$$

i.e.

$$\boxed{f_{\text{RS}}(\beta) = -\frac{\beta J^2}{4} (1 - q)^2 - \frac{1}{\beta} \int \mathcal{D}z \ln 2 \cosh (\beta J \sqrt{q} z).} \quad (2.7)$$

Stationarity reproduces the self-consistency equation. Using $\partial_q \Xi = \beta J z / (2\sqrt{q})$ and Gaussian integration by parts, $\int \mathcal{D}z z g(z) = \int \mathcal{D}z g'(z)$,

$$\frac{\partial}{\partial q} \int \mathcal{D}z \ln 2 \cosh \Xi = \frac{\beta J}{2\sqrt{q}} \int \mathcal{D}z z \tanh \Xi = \frac{(\beta J)^2}{2} \int \mathcal{D}z \text{sech}^2 \Xi = \frac{(\beta J)^2}{2} \left(1 - \int \mathcal{D}z \tanh^2 \Xi\right), \quad (2.8)$$

so $\partial f_{\text{RS}}/\partial q = 0$ gives $q = \int \mathcal{D}z \tanh^2(\beta J \sqrt{q} z)$, as in Eq. (2.3). Linearizing, $q = (\beta J)^2 q + O(q^2)$, so a nonzero solution appears continuously at $T_c = J$. Just below, $q \simeq \tau + O(\tau^2)$ with $\tau = (T_c - T)/T_c$. With a field the same steps give $\Xi = \beta(J\sqrt{q}z + h)$, the form used in the AT analysis below.

3 The fluctuation Hessian

Expand the exponent of Eq. (1.4) to second order in $\delta Q_{ab} = Q_{ab} - q$. Write the exponent per site as $G(Q) = -\frac{\beta^2}{2} \sum_{a < b} Q_{ab}^2 + \ln \text{Tr} e^{L(Q)}$ and use $\partial \ln \text{Tr} e^L / \partial Q_{ab} = \beta^2 \langle s^a s^b \rangle$:

$$-\frac{\partial^2 G}{\partial Q_{ab} \partial Q_{cd}} = \beta^2 \left[\delta_{(ab),(cd)} - \beta^2 (\langle s^a s^b s^c s^d \rangle - \langle s^a s^b \rangle \langle s^c s^d \rangle) \right] \equiv \beta^2 \mathbf{G}_{(ab),(cd)}. \quad (3.1)$$

Stability of the RS saddle requires $\mathbf{G}$ to be positive semidefinite on the physical fluctuation space. By the permutation symmetry of the RS saddle, $\mathbf{G}_{(ab),(cd)}$ depends only on how many replica indices the two pairs share, so it takes exactly three values:

$$\begin{aligned} M_1 &\equiv \mathbf{G}_{(ab),(ab)} = 1 - \beta^2(1 - q^2) && \text{(two shared indices; } (s^a s^b)^2 = 1), \\ M_2 &\equiv \mathbf{G}_{(ab),(ac)} = -\beta^2(q - q^2) && \text{(one shared index; } s^a s^b s^a s^c = s^b s^c), \\ M_3 &\equiv \mathbf{G}_{(ab),(cd)} = -\beta^2(r - q^2) && \text{(no shared index).} \end{aligned} \quad (3.2)$$

The matrix acts on the $R(R-1)/2$ -dimensional space of pairs. It can be diagonalized exactly for any R using the permutation symmetry.

4 Exact diagonalization by symmetry

The three invariant subspaces

$\mathbf{G}$ commutes with the action of the permutation group S_R on pair indices, so the pair space decomposes into irreducible S_R -representations. There are exactly three.

Longitudinal (dimension 1). The uniform vector $v_{ab} = 1$. The pair (ab) has $2(R-2)$ partners sharing one index (choose the shared index in 2 ways and the other index in $R-2$ ways) and $\binom{R-2}{2}$ disjoint partners. Hence

$$\lambda_L = M_1 + 2(R-2)M_2 + \frac{(R-2)(R-3)}{2}M_3. \quad (4.1)$$

Anomalous (dimension $R-1$). Vectors of the form $v_{ab} = x_a + x_b$ with $\sum_a x_a = 0$. These single out individual replicas within the pair structure. Acting with $\mathbf{G}$,

$$\begin{aligned} (\mathbf{G}v)_{ab} &= M_1(x_a + x_b) + M_2 \sum_{c \neq a,b} [(x_a + x_c) + (x_b + x_c)] + M_3 \sum_{\substack{c < d \\ \{c,d\} \cap \{a,b\} = \emptyset}} (x_c + x_d) \\ &= M_1(x_a + x_b) + M_2 \left[(R-2)(x_a + x_b) + 2 \sum_{c \neq a,b} x_c \right] + M_3(R-3) \sum_{c \neq a,b} x_c. \end{aligned} \quad (4.2)$$

Using $\sum_{c \neq a,b} x_c = -(x_a + x_b)$ from the constraint,

$$\lambda_A = M_1 + (R-4)M_2 - (R-3)M_3. \quad (4.3)$$

The family $\{x_a + x_b\}$ with unconstrained x has dimension R (for $R \geq 3$ the map $x \mapsto \{x_a + x_b\}$ is injective); removing the uniform direction ($x_a = \text{const}$, which gives the longitudinal vector) leaves $R-1$ anomalous directions.

Replicon (dimension $R(R-3)/2$). Everything orthogonal to the span of $\{x_a + x_b\}$. Orthogonality is equivalent to vanishing row sums:

$$\sum_{\text{pairs}} \eta_{ab}(x_a + x_b) = \sum_a x_a \sum_{b \neq a} \eta_{ab} = 0 \quad \forall x \quad \Longleftrightarrow \quad \sum_{b \neq a} \eta_{ab} = 0 \quad \forall a. \quad (4.4)$$

These are R independent linear constraints, so

$$\dim(\text{replicon}) = \frac{R(R-1)}{2} - R = \frac{R(R-3)}{2}. \quad (4.5)$$

Acting with $\mathbf{G}$ on such an η ,

$$(\mathbf{G}\eta)_{ab} = M_1 \eta_{ab} + M_2 \sum_{c \neq a, b} (\eta_{ac} + \eta_{bc}) + M_3 \sum_{\substack{c < d \\ \text{disjoint from } (ab)}} \eta_{cd}. \quad (4.6)$$

The row-sum constraints give $\sum_{c \neq a, b} \eta_{ac} = -\eta_{ab}$, and likewise for b , so the M_2 term is $-2M_2 \eta_{ab}$. The sum of all components vanishes (sum the row constraints), so the disjoint sum equals $0 - \eta_{ab} - (\text{one-shared sum}) = -\eta_{ab} + 2\eta_{ab} = +\eta_{ab}$. Therefore

$$\boxed{\lambda_R = M_1 - 2M_2 + M_3} \quad (4.7)$$

with multiplicity $R(R-3)/2$: zero at $R = 3$, and first nontrivial (dimension two) at $R = 4$; at $R = 2$ the pair space is one-dimensional and purely longitudinal. The alternating combination in Eq. (4.7) is a connected part: it removes from the pair fluctuation everything that can be attributed to single replicas.

Consistency check and a shortcut

Trace check. The eigenvalues must satisfy $\text{Tr } \mathbf{G} = \frac{R(R-1)}{2} M_1 = \lambda_L + (R-1)\lambda_A + \frac{R(R-3)}{2} \lambda_R$. The M_1 coefficients sum to $1 + (R-1) + R(R-3)/2 = R(R-1)/2$; the M_2 coefficients to $2(R-2) + (R-1)(R-4) - R(R-3) = 0$; the M_3 coefficients to $(R-3)[\frac{R-2}{2} - (R-1) + \frac{R}{2}] = 0$.

Johnson scheme. Write $\mathbf{G} = M_1 \mathbb{I} + M_2 A + M_3 D$, where A is the adjacency matrix of the triangular (Johnson) graph on pairs (two pairs adjacent if they share exactly one index) and $D = \mathbf{J} - \mathbb{I} - A$, with $\mathbf{J}$ the all-ones matrix. The eigenvalues of A are $2(R-2)$, $R-4$ and -2 with multiplicities 1, $R-1$ and $R(R-3)/2$. Substituting reproduces Eqs. (4.1)–(4.7).

The replica limit

At $R \rightarrow 0$ the longitudinal and anomalous eigenvalues become degenerate,

$$\lambda_L, \lambda_A \rightarrow M_1 - 4M_2 + 3M_3, \quad \lambda_R \rightarrow M_1 - 2M_2 + M_3, \quad (4.8)$$

while the multiplicities continue analytically to 1, -1 and 0 (summing to $R(R-1)/2 \rightarrow 0$, as they must). Negative and vanishing multiplicities are a consequence of the analytic continuation; the eigenvalues themselves remain meaningful and control the propagators of the replica field theory.

5 The de Almeida–Thouless instability

Insert the SK matrix elements (3.2) into Eq. (4.7):

$$\lambda_R = 1 - \beta^2 \left[(1 - q^2) - 2(q - q^2) + (r - q^2) \right] = 1 - \beta^2 (1 - 2q + r). \quad (5.1)$$

With Eq. (2.3), $1 - 2q + r = \int \mathcal{D}z (1 - \tanh^2 \Xi)^2$, so

$$\lambda_R = 1 - \beta^2 \int \mathcal{D}z \operatorname{sech}^4 [\beta(\sqrt{q}z + h)]. \quad (5.2)$$

The AT condition is $\lambda_R = 0$ (de Almeida and Thouless, 1978).

Above T_c at $h = 0$. Here $q = r = 0$ and $\lambda_L = \lambda_A = \lambda_R = 1 - \beta^2$: all three sectors soften together at $T_c = 1$. The transition itself is not an RSB phenomenon.

Below T_c at $h = 0$. Expanding Eq. (5.2) with the RS solution $q \simeq \tau + O(\tau^2)$, the $O(\tau)$ terms cancel and

$$\lambda_R = -\frac{4}{3} \tau^2 < 0 \quad (5.3)$$

(Exercise 3). The RS solution of SK is unstable immediately below T_c , and the instability lies entirely in the replicon sector (λ_L, λ_A remain positive).

In a field. For $h \neq 0$ there is no ordering transition, but Eq. (5.2) still changes sign on the AT line $h_{\text{AT}}(T)$; near T_c ,

$$\left(\frac{h_{\text{AT}}}{J} \right)^2 \simeq \frac{4}{3} \tau^3. \quad (5.4)$$

Below the line replica symmetry is broken (Parisi, 1979); in the full RSB solution the replicon becomes marginal, $\lambda_R = 0$, rather than negative.

6 Negative entropy of the RS solution

The AT computation shows which mode is unstable; historically, the first sign that the RS solution is wrong was thermodynamic (Sherrington and Kirkpatrick, 1975). We now extract the low-temperature behaviour of Eq. (2.7) (J restored, $h = 0$).

Low-temperature order parameter. The self-consistency equation can be written $1 - q = \int \mathcal{D}z \operatorname{sech}^2(\beta J \sqrt{q} z)$. As $\beta \rightarrow \infty$ the integrand is a peak of unit height and width $\sim 1/(\beta J)$ at $z = 0$, so the Gaussian weight can be replaced by its value at the origin:

$$\int \mathcal{D}z \operatorname{sech}^2(\beta J \sqrt{q} z) = \frac{1}{\sqrt{2\pi}} \frac{1}{\beta J \sqrt{q}} \int_{-\infty}^{\infty} du \operatorname{sech}^2 u + O(\beta^{-3}) = \sqrt{\frac{2}{\pi}} \frac{T}{J \sqrt{q}} + O(T^3), \quad (6.1)$$

using $\int \operatorname{sech}^2 u du = 2$. With $q \rightarrow 1$,

$$q \simeq 1 - aT, \quad a = \sqrt{\frac{2}{\pi}} \frac{1}{J}. \quad (6.2)$$

Low-temperature free energy. Insert this into Eq. (2.7), term by term. The first term is quadratic in $(1 - q)$:

$$-\frac{\beta J^2}{4}(1 - q)^2 = -\frac{J^2 a^2}{4}T = -\frac{T}{2\pi}, \quad (6.3)$$

since $J^2 a^2 = 2/\pi$. For the second term use $\ln 2 \cosh x = |x| + \ln(1 + e^{-2|x|})$:

$$-\frac{1}{\beta} \int \mathcal{D}z \ln 2 \cosh(\beta J \sqrt{q} z) = -J \sqrt{q} \int \mathcal{D}z |z| - 2T \int_0^\infty \mathcal{D}z \ln(1 + e^{-2\beta J \sqrt{q} z}). \quad (6.4)$$

The first piece, with $\int \mathcal{D}z |z| = \sqrt{2/\pi}$ and $\sqrt{q} \simeq 1 - aT/2$, gives

$$-J \sqrt{q} \sqrt{\frac{2}{\pi}} = -\sqrt{\frac{2}{\pi}} J + \sqrt{\frac{2}{\pi}} \frac{Ja}{2} T = -\sqrt{\frac{2}{\pi}} J + \frac{T}{\pi}. \quad (6.5)$$

The second piece is $O(T^2)$: substituting $u = 2\beta J \sqrt{q} z$ and replacing the Gaussian by its value at the origin,

$$-2T \int_0^\infty \mathcal{D}z \ln(1 + e^{-2\beta J \sqrt{q} z}) \simeq -\frac{2T}{\sqrt{2\pi}} \frac{1}{2\beta J} \int_0^\infty du \ln(1 + e^{-u}) = -\frac{\pi^2 T^2}{12\sqrt{2\pi} J}, \quad (6.6)$$

using $\int_0^\infty \ln(1 + e^{-u}) du = \pi^2/12$. Collecting,

$$f_{\text{RS}}(T) = -\sqrt{\frac{2}{\pi}} J + \frac{T}{\pi} - \frac{T}{2\pi} + O(T^2) = -\sqrt{\frac{2}{\pi}} J + \frac{T}{2\pi} + O(T^2). \quad (6.7)$$

The contradiction. The entropy $S = -\partial f / \partial T$ therefore approaches

$$\boxed{S(0) = -\frac{1}{2\pi} \approx -0.159,} \quad (6.8)$$

which is impossible: a system of discrete spins has $S \geq 0$. The RS ground-state energy, $e_{\text{RS}} = -\sqrt{2/\pi} J \approx -0.798 J$, likewise lies below the exact (Parisi) value $-0.7633 J$. Even the paramagnetic branch shows the problem: continuing the $q = 0$ solution $f = -\beta J^2/4 - T \ln 2$ below T_c gives $S = \ln 2 - J^2/4T^2$, which is negative for $T < J/(2\sqrt{\ln 2}) \approx 0.60 J$.

The negative entropy is the thermodynamic symptom; the AT computation of the previous section is the mechanism. At $h = 0$ the replicon eigenvalue is negative for all $T < T_c$, so the RS saddle is the wrong one throughout the low-temperature phase, and the error grows toward $T = 0$. In Parisi's full-RSB solution the entropy is $S(0) = 0$, $e_0 \simeq -0.7633 J$, and the replicon is exactly marginal, $\lambda_R = 0$, throughout the glass phase. The smallness of the discrepancy (-0.16 compared with the paramagnetic $\ln 2 \approx 0.69$) explains why the RS solution looks reasonable at first sight.

7 Saddle points in the replica limit: why “extremize”?

The free energy is written as an extremum of the effective action rather than a maximum. For a genuine Laplace integral one should take the maximum of the exponent; the difference comes from the continuation $n \rightarrow 0$. A simple toy example (Mézard and Montanari, 2009) shows this. Consider

$$Z_{\text{toy}}(n) = \left(\frac{2\pi}{N}\right)^{n(n-1)/4}, \quad (7.1)$$

defined for real n . For integer n , the Gaussian identity $\sqrt{2\pi/N} = \int e^{-Nz^2/2} dz$, applied once per unordered pair $(\alpha\beta)$, gives the exact integral representation

$$Z_{\text{toy}}(n) = \int \prod_{(\alpha\beta)} dQ_{\alpha\beta} \exp\left(-\frac{N}{2} \sum_{(\alpha\beta)} Q_{\alpha\beta}^2\right) \equiv \int \prod_{(\alpha\beta)} dQ_{\alpha\beta} e^{Ng(Q)}, \quad (7.2)$$

with $n(n-1)/2$ integration variables. On the replica-symmetric line $Q_{\alpha\beta} = q_0$,

$$g(q_0) = -\frac{n(n-1)}{4}q_0^2, \quad g''(q_0) = -\frac{n(n-1)}{2}. \quad (7.3)$$

For $n > 1$ the stationary point $q_0 = 0$ is a maximum of g , as a saddle-point evaluation requires. For $0 < n < 1$, however, $g''(0) = -n(n-1)/2 > 0$: the same point is now a minimum along the replica-symmetric direction, and it is still the correct saddle, since $g(0) = 0$ reproduces the exact Z_{toy} to exponential accuracy. The analytic continuation in n changes the character of the stationary points; the number of transverse directions, $n(n-1)/2$, itself becomes negative for $0 < n < 1$, so “maximum” is not a meaningful requirement. The working prescription, used throughout these notes, is: find the stationary points and check stability sector by sector (longitudinal, anomalous, replicon) with the eigenvalues continued in n . The same inversion explains a well-known feature of the SK solution: in the replica limit the free energy must be maximized, not minimized, over the variational parameters of the RSB ansatz below.

8 Replica-symmetry breaking

The negative entropy and the negative replicon eigenvalue express the same fact: below T_c the true saddle lies outside the replica-symmetric subspace. Parisi’s strategy ([Parisi, 1979, 1980](#)) is to enlarge the variational family recursively, in a way that keeps the single-site trace computable. In this section J is restored and $h = 0$.

The 1RSB ansatz and its free energy

For integer replica number R , choose an integer $m_1 \leq R$ dividing R , split the replicas into R/m_1 consecutive blocks of size m_1 , and set

$$Q_{ab} = \begin{cases} q_1, & a \neq b \text{ in the same block,} \\ q_0, & a, b \text{ in different blocks.} \end{cases} \quad (8.1)$$

The RS ansatz is recovered for $m_1 = 1$ or $q_1 = q_0$. Two counting identities are needed. First, the interaction term of the single-site problem can be written with squares of block magnetizations,

$$\sum_{a < b} Q_{ab} s^a s^b = \frac{1}{2} \left[q_0 \left(\sum_a s^a \right)^2 + (q_1 - q_0) \sum_B \left(\sum_{a \in B} s^a \right)^2 - Rq_1 \right], \quad (8.2)$$

where B runs over the R/m_1 blocks (the last term removes the diagonal $a = b$ terms, which contribute $q_0 R + (q_1 - q_0) R = q_1 R$). Second, counting pairs,

$$\sum_{a < b} Q_{ab}^2 = \frac{R(R - m_1)}{2} q_0^2 + \frac{R(m_1 - 1)}{2} q_1^2, \quad (8.3)$$

since each replica has $R - m_1$ partners outside its block and $m_1 - 1$ inside. Linearize the two squares with two Hubbard–Stratonovich fields, a global field z_0 for $(\sum_a s^a)^2$ and one field z_B per block:

$$\text{Tr} e^{(\beta J)^2 \sum_{a < b} Q_{ab} s^a s^b} = e^{-(\beta J)^2 R q_1 / 2} \int \mathcal{D} z_0 \left[\int \mathcal{D} z_1 (2 \cosh \Xi(z_0, z_1))^{m_1} \right]^{R/m_1}, \quad (8.4)$$

$$\Xi(z_0, z_1) = \beta J (\sqrt{q_0} z_0 + \sqrt{q_1 - q_0} z_1), \quad (8.5)$$

because at fixed (z_0, z_B) the spins decouple, each block contributes $[2 \cosh \Xi]^{m_1}$, and the R/m_1 blocks are identical. Now assemble the action as in the RS case, expand $[\dots]^{R/m_1} = 1 + \frac{R}{m_1} \ln[\dots] + O(R^2)$, divide by R and take $R \rightarrow 0$. The pair counts per replica, $(R - m_1)/2 \rightarrow -m_1/2$ and $(m_1 - 1)/2$, give

$$-\beta f_{1\text{RSB}} = \frac{(\beta J)^2}{4} \left[1 - 2q_1 + m_1 q_0^2 - (m_1 - 1) q_1^2 \right] + \frac{1}{m_1} \int \mathcal{D} z_0 \ln \int \mathcal{D} z_1 (2 \cosh \Xi)^{m_1}. \quad (8.6)$$

Checks: for $q_1 = q_0$ the z_1 integral drops out and Eq. (8.6) reduces to the RS free energy (2.7); the same happens for $m_1 = 1$. In the replica limit the block size continues to $0 \leq m_1 \leq 1$ and becomes a variational parameter.

Stationarity conditions

Define the weighted average within one block

$$\langle g \rangle_* \equiv \frac{\int \mathcal{D} z_1 [2 \cosh \Xi]^{m_1} g(\Xi)}{\int \mathcal{D} z_1 [2 \cosh \Xi]^{m_1}}. \quad (8.7)$$

Differentiating the last term of Eq. (8.6) with respect to q_1 , using $\partial_{q_1} \Xi = \beta J z_1 / (2\sqrt{q_1 - q_0})$ and one Gaussian integration by parts in z_1 ,

$$\frac{\partial}{\partial q_1} \frac{1}{m_1} \int \mathcal{D} z_0 \ln \int \mathcal{D} z_1 [2 \cosh \Xi]^{m_1} = \frac{(\beta J)^2}{2} \left[1 + (m_1 - 1) \int \mathcal{D} z_0 \langle \tanh^2 \Xi \rangle_* \right], \quad (8.8)$$

where $\text{sech}^2 = 1 - \tanh^2$ was used, while the polynomial part contributes $-\frac{(\beta J)^2}{2} [1 + (m_1 - 1) q_1]$. Hence

$$q_1 = \int \mathcal{D} z_0 \langle \tanh^2 \Xi \rangle_*. \quad (8.9)$$

The same computation for q_0 (one integration by parts in z_0 , one in z_1 ; the first produces a square of the weighted average) gives

$$q_0 = \int \mathcal{D} z_0 \langle \tanh \Xi \rangle_*^2, \quad (8.10)$$

and stationarity in m_1 gives the transcendental condition

$$\frac{(\beta J)^2}{4} (q_0^2 - q_1^2) - \frac{1}{m_1^2} \int \mathcal{D} z_0 \ln W(z_0) + \frac{1}{m_1} \int \mathcal{D} z_0 \langle \ln 2 \cosh \Xi \rangle_* = 0, \quad W = \int \mathcal{D} z_1 [2 \cosh \Xi]^{m_1}. \quad (8.11)$$

Solving the three equations numerically below T_c , 1RSB removes most of the RS error: the zero-temperature entropy improves from -0.159 to about -0.01 , and the ground-state energy comes within a fraction of a percent of the exact value. Repeating the stability analysis sector by sector on the 1RSB saddle shows, however, that it is still unstable (the replicon-type eigenvalue within the innermost blocks remains negative), so the construction must be iterated.

k RSB and the Parisi limit

At step k , choose block sizes $m_1 > m_2 > \dots > m_k$ (each dividing the previous), let the off-diagonal blocks at depth j carry the value q_j , and let the innermost blocks repeat the $(k-1)$ RSB pattern. Each level adds one nested Gaussian field, so the single-site trace becomes a k -fold nested integral of the 1RSB type. In the replica limit the order of the block sizes is reversed, $0 \leq m_1 \leq m_2 \leq \dots \leq m_k \leq 1$, and all q_j, m_j are variational. As $k \rightarrow \infty$ the step function $q(x)$, defined by $q(x) = q_j$ for $x \in [m_j, m_{j+1})$, becomes a continuous nondecreasing function on $[0, 1]$: this is Parisi's full-RSB solution (Parisi, 1979, 1980; Mézard et al., 1987). Its two key properties, stated here without derivation: the zero-temperature entropy vanishes, removing the contradiction of the RS solution exactly, with ground-state energy $e_0 \simeq -0.7633 J$; and the replicon eigenvalue is exactly marginal, $\lambda_R = 0$, throughout the glass phase. The spin-glass phase is permanently at the edge of stability, which is the origin of its slow dynamics and its divergent spin-glass susceptibility.

9 The overlap distribution and ultrametricity

The physical meaning of replica-symmetry breaking is contained in the overlap between two real copies of the same sample. For two independent equilibrium configurations s^1, s^2 with the same couplings J ,

$$\hat{q}_{12} = \frac{1}{N} \sum_i s_i^1 s_i^2, \quad P_J(q) = \text{Tr}_{s^1, s^2} \left[\frac{e^{-\beta H[s^1]} e^{-\beta H[s^2]}}{Z_J^2} \delta(\hat{q}_{12} - q) \right]. \quad (9.1)$$

The disorder average $\mathbb{E}[P_J(q)]$ is computed by a small modification of the replica argument. The difficulty is the explicit factor Z_J^{-2} ; multiply numerator and denominator by Z_J^{n-2} and let $n \rightarrow 0$, so that the denominator $Z_J^n \rightarrow 1$ inside the disorder average:

$$\mathbb{E}[P_J(q)] = \lim_{n \rightarrow 0} \mathbb{E} \left[\text{Tr}_{s^1, \dots, s^n} \prod_{a=1}^n e^{-\beta H[s^a]} \delta \left(\frac{1}{N} \sum_i s_i^1 s_i^2 - q \right) \right]. \quad (9.2)$$

The disorder average and Hubbard–Stratonovich steps go through as before, and in the thermodynamic limit the δ -function selects the saddle-point value of Q_{12} . Since every permutation of a saddle is again a saddle with the same action, one must sum over the S_n orbit, i.e. average over the matrix elements:

$$\mathbb{E}[P_J(q)] = \lim_{n \rightarrow 0} \frac{1}{n(n-1)} \sum_{a \neq b} \delta(Q_{ab}^* - q). \quad (9.3)$$

The overlap distribution is the distribution of the off-diagonal elements of the saddle-point matrix.

RS. All elements equal q_0 : $P(q) = \delta(q - q_0)$. There is a single pure state (up to the global spin flip), and the overlap does not fluctuate.

1RSB. A fraction $(m_1 - 1)/(n - 1)$ of the pairs is intra-block; in the limit $n \rightarrow 0$ this fraction becomes $1 - m_1$, and

$$P(q) = m_1 \delta(q - q_0) + (1 - m_1) \delta(q - q_1). \quad (9.4)$$

Normalization and positivity of these weights is exactly the condition $0 \leq m_1 \leq 1$ found above: the block size becomes a probability. Two equilibrium copies either fall into the same valley (overlap q_1 , probability $1 - m_1$) or into different valleys (overlap q_0). At k RSB, $P(q)$ is a sum of $k + 1$ delta

functions; in the Parisi limit it acquires a continuous part, with $x(q) = \int_0^q P(q') dq'$ the inverse of the order-parameter function $q(x)$. A nontrivial $P(q)$ means many pure states separated by free-energy barriers that diverge with N : replica-symmetry breaking is ergodicity breaking.

Ultrametricity. For three copies the same argument gives

$$\mathbb{E}[P_J(q_{12}, q_{23}, q_{31})] = \lim_{n \rightarrow 0} \frac{1}{n(n-1)(n-2)} \sum_{a \neq b \neq c \neq a} \delta(Q_{ab}^* - q_{12}) \delta(Q_{bc}^* - q_{23}) \delta(Q_{ca}^* - q_{31}). \quad (9.5)$$

For a hierarchical (Parisi) matrix, Q_{ab} depends only on the smallest block containing both a and b . Take any triple and the smallest block containing all three. Its sub-blocks either separate all three replicas, in which case the three overlaps are equal, or group two of them (say a, b) apart from c , in which case $q_{ac} = q_{bc} < q_{ab}$. In both cases the two smallest overlaps are equal: if $q_1 \geq q_2 \geq q_3$ then $q_2 = q_3$; equivalently $q_{ab} \geq \min(q_{ac}, q_{cb})$ for all triples. In the metric $d_{ab}^2 = 2(q_{EA} - q_{ab})$ this is the ultrametric inequality: the pure states of the SK model are organized as the leaves of a tree (Mézard et al., 1984).

On the Nishimori line (last section) the gauge identities force the overlap distribution to coincide with the magnetization distribution, $P(q) = P(m)$, so no ultrametric structure can exist there; this is the same conclusion as $\chi_{SG} = \chi_F$, at the level of distributions.

10 Physical meaning of the three sectors

Longitudinal. The uniform mode $\delta Q_{ab} = \delta q$ is the fluctuation of the magnitude of the typical overlap. It mixes with energy and magnetization fluctuations and softens at ordinary thermodynamic transitions.

Anomalous. The modes $x_a + x_b$ single out one replica: they are one-replica responses expressed through the pair field, degenerate with the longitudinal sector in the replica limit.

Replicon. The row-traceless modes are genuinely two-replica objects: they cannot be decomposed into single-replica fields. Their propagator determines the spin-glass susceptibility (Bray and Moore, 1979),

$$\chi_{SG} = \frac{1}{N} \sum_{ij} \mathbb{E}[\langle s_i s_j \rangle_c^2] \sim \frac{1}{\lambda_R}, \quad (10.1)$$

i.e. λ_R is the mass of the overlap channel: it measures how strongly correlations differ from sample to sample. When it vanishes, these sample-to-sample fluctuations of correlations become long-ranged, the overlap distribution broadens, and replica symmetry breaks. This is why the replicon is associated with RSB: its softening is the RSB instability. The sector itself, however, exists in any replica-symmetric theory: it follows from the representation theory, whether or not the instability occurs.

11 The Nishimori line: where the instability cannot occur

Consider a system whose quenched disorder is drawn from a distribution matched to the Boltzmann weight: the random-bond Ising model (RBIM) with $J_{ij} = \pm 1$, $P(J_{ij} = -1) = p$, at inverse

temperature β (Nishimori, 1981). Write the disorder distribution in Gibbs form,

$$P(J_{ij}) = \frac{e^{\beta_p J_{ij}}}{2 \cosh \beta_p}, \quad \tanh \beta_p = 1 - 2p. \quad (11.1)$$

The Nishimori line (NL) is $\beta = \beta_p$.

Gauge identity. Under the gauge transformation $J_{ij} \rightarrow J_{ij} \sigma_i \sigma_j$, $s_i \rightarrow s_i \sigma_i$ (any $\sigma_i = \pm 1$), the Hamiltonian is invariant, $\langle s_0 s_r \rangle_{\beta, J^\sigma} = \sigma_0 \sigma_r \langle s_0 s_r \rangle_{\beta, J}$, and the disorder weight picks up the gauge factor. Averaging over all 2^N gauge choices (N_B is the number of bonds),

$$\begin{aligned} \mathbb{E}[\langle s_0 s_r \rangle_\beta] &= \sum_J \frac{e^{\beta_p \sum J}}{(2 \cosh \beta_p)^{N_B}} \langle s_0 s_r \rangle_{\beta, J} = \frac{1}{2^N} \sum_\sigma \sum_J \frac{e^{\beta_p \sum J^\sigma}}{(2 \cosh \beta_p)^{N_B}} \sigma_0 \sigma_r \langle s_0 s_r \rangle_{\beta, J} \\ &= \sum_J \frac{Z_{\beta_p}(J)}{2^N (2 \cosh \beta_p)^{N_B}} \langle \sigma_0 \sigma_r \rangle_{\beta_p, J} \langle s_0 s_r \rangle_{\beta, J}, \end{aligned} \quad (11.2)$$

where in the first step we relabelled $J \rightarrow J^\sigma$ inside the sum, and in the second we recognized $\sum_\sigma \sigma_0 \sigma_r e^{\beta_p \sum J^\sigma} = Z_{\beta_p}(J) \langle \sigma_0 \sigma_r \rangle_{\beta_p, J}$. The same manipulation applied to the constant function shows that $Z_{\beta_p}(J)/[2^N (2 \cosh \beta_p)^{N_B}]$ is again the disorder measure (Exercise 4), so

$$\mathbb{E}[\langle s_0 s_r \rangle_\beta] = \mathbb{E}[\langle s_0 s_r \rangle_{\beta_p} \langle s_0 s_r \rangle_\beta] \xrightarrow{\beta=\beta_p} \mathbb{E}[\langle s_0 s_r \rangle] = \mathbb{E}[\langle s_0 s_r \rangle^2]. \quad (11.3)$$

Consequence for the replicon. Summing Eq. (11.3) over r (in the phase where the susceptibilities are finite),

$$\boxed{\chi_F = \sum_r \mathbb{E} \langle s_0 s_r \rangle = \sum_r \mathbb{E} \langle s_0 s_r \rangle^2 = \chi_{SG}.} \quad (11.4)$$

The spin-glass susceptibility, i.e. the replicon-channel correlator, is identically equal to the ferromagnetic one on the NL. The replicon mass is therefore fixed to the mass of the ordering channel: it cannot vanish where the ferromagnetic channel is massive, no AT line crosses the NL, and there is no replica-symmetry breaking on it (Nishimori, 2001). Overlap and ordering fluctuations soften together, at the multicritical (Nishimori) point where the paramagnetic, ferromagnetic and glassy phase boundaries meet. In replica language the NL has an enlarged permutation symmetry S_{R+1} : the planted (true) configuration joins the R replicas as an equivalent member, so the truth-sample and sample-sample overlaps have the same statistics; this is the symmetry form of Eq. (11.3). Away from the line the identity fails, the replicon can soften independently, and glassy phases become possible.

Exercises

Exercise 1. Verify the trace identity $\frac{R(R-1)}{2} M_1 = \lambda_L + (R-1)\lambda_A + \frac{R(R-3)}{2} \lambda_R$ directly from Eqs. (4.1)–(4.7).

Exercise 2. Show that at $R = 4$ the six pairs decompose as $6 = 1 + 3 + 2$, and construct the two replicon vectors explicitly, e.g. $\eta = Q_{12} + Q_{34} - Q_{13} - Q_{24}$ and $\eta' = Q_{12} + Q_{34} - Q_{14} - Q_{23}$; check the row-sum constraints (4.4) and $G\eta = (M_1 - 2M_2 + M_3)\eta$.

Exercise 3. Expand Eq. (5.2) at $h = 0$ to second order in $\tau = (T_c - T)/T_c$, using $\tanh^2 x = x^2 - \frac{2}{3}x^4 + \frac{17}{45}x^6 - \dots$ and the self-consistent $q(\tau)$ to the same order, and determine the coefficient c in $\lambda_R = -c\tau^2$. Then restore $h \neq 0$ and derive the AT line $h_{\text{AT}}^2 \simeq \frac{4}{3}\tau^3$.

Exercise 4. Prove that the gauge-averaged measure appearing above Eq. (11.3) is normalized, i.e. $\sum_J Z_{\beta_p}(J)/[2^N(2 \cosh \beta_p)^{N_B}] = 1$.

Exercise 5. On the NL, derive the internal-energy identity $\mathbb{E}[\langle J_{ij}s_i s_j \rangle] = \tanh \beta_p$ by the same gauge argument, and conclude that the internal energy is analytic across the Nishimori point.

Exercise 6. Verify $\int_0^\infty \ln(1 + e^{-u}) du = \pi^2/12$ (expand the logarithm and sum the alternating series), and confirm that the finite- T correction to the second term of the RS free energy at low T is $O(T^2)$, so that it does not contribute to $S(0)$.

Exercise 7. From the RS self-consistency equation, derive the behaviour of $q(\tau)$ near T_c to second order in $\tau = (T_c - T)/T_c$, and verify $T_c = J$.

Exercise 8. Show from Eq. (9.3) that the 1RSB overlap distribution is $P(q) = m_1 \delta(q - q_0) + (1 - m_1) \delta(q - q_1)$, and that normalization and positivity of the weights require $0 \leq m_1 \leq 1$ in the replica limit.

Solutions

Solution 1 (trace identity). $\text{Tr } \mathbf{G}$ counts M_1 once per pair, i.e. $\frac{R(R-1)}{2} M_1$, and must equal $\lambda_L + (R-1)\lambda_A + \frac{R(R-3)}{2} \lambda_R$. Collect coefficients from Eqs. (4.1)–(4.7):

$$\begin{aligned} M_1 : & 1 + (R-1) + \frac{R(R-3)}{2} = \frac{2+2R-2+R^2-3R}{2} = \frac{R(R-1)}{2}, \\ M_2 : & 2(R-2) + (R-1)(R-4) - R(R-3) = (2R-4) + (R^2-5R+4) - (R^2-3R) = 0, \\ M_3 : & \frac{(R-2)(R-3)}{2} - (R-1)(R-3) + \frac{R(R-3)}{2} = (R-3) \left[\frac{R-2}{2} - (R-1) + \frac{R}{2} \right] = 0. \quad \square \end{aligned}$$

Solution 2 (replicon vectors at $R = 4$). The six pairs split as $6 = 1 + 3 + 2$. Take $\eta = Q_{12} + Q_{34} - Q_{13} - Q_{24}$. Row sums: replica 1 appears in pairs (12), (13), (14) with coefficients $+1, -1, 0$, sum 0; replica 2 in (12), (23), (24) with $+1, 0, -1$, sum 0; replicas 3 and 4 likewise. The vector $\eta' = Q_{12} + Q_{34} - Q_{14} - Q_{23}$ passes the same check, and η, η' are linearly independent; together they span the $R(R-3)/2 = 2$ replicon directions. The (12) component of $\mathbf{G}\eta$ is $M_1(+1) + M_2[\eta_{13} + \eta_{23} + \eta_{14} + \eta_{24}] + M_3\eta_{34} = M_1 + M_2(-1+0+0-1) + M_3(+1) = M_1 - 2M_2 + M_3$, and by symmetry $\mathbf{G}\eta = \lambda_R \eta$ componentwise. $\square$

Solution 3 (AT coefficient and AT line). Work at $J = 1$, $\varepsilon \equiv \beta^2 - 1$. Gaussian moments are $\mathbb{E}z^2 = 1$, $\mathbb{E}z^4 = 3$, $\mathbb{E}z^6 = 15$, and $\tanh^2 x = x^2 - \frac{2}{3}x^4 + \frac{17}{45}x^6 - \dots$, $\tanh^4 x = x^4 - \frac{4}{3}x^6 + \dots$. With $\Xi = \beta\sqrt{q}z$ ($h = 0$) the self-consistency equation becomes

$$q = \beta^2 q - 2\beta^4 q^2 + \frac{17}{3}\beta^6 q^3 - \dots \implies q = \frac{\varepsilon}{2} - \frac{7}{24}\varepsilon^2 + O(\varepsilon^3).$$

Using $\mathbb{E} \tanh^2 \Xi = q$ (the saddle equation) and $\mathbb{E} \tanh^4 \Xi = 3\beta^4 q^2 - 20\beta^6 q^3 + \dots$,

$$\lambda_R = 1 - \beta^2(1 - 2q + \mathbb{E} \tanh^4 \Xi) = -\varepsilon + 2\beta^2 q - 3\beta^6 q^2 + O(\varepsilon^3) = -\frac{\varepsilon^2}{3} + O(\varepsilon^3).$$

With $\varepsilon \simeq 2\tau$ this is $\lambda_R \simeq -\frac{4}{3}\tau^2$, so $c = 4/3$. With a small field, power counting gives $q \sim \tau$, $h^2 \sim \tau^3$; the field enters the q -equation as $+\beta^2 h^2$ and $\mathbb{E} \tanh^4$ only at higher order. Eliminating ε via the q -equation, $\varepsilon = 2\beta^4 q - \frac{17}{3}\beta^6 q^2 - \beta^2 h^2/q$, and substituting into λ_R gives

$$\lambda_R \simeq -\frac{\varepsilon^2}{3} + \frac{h^2}{q}, \quad \lambda_R = 0 \implies h_{\text{AT}}^2 = \frac{q\varepsilon^2}{3} = \frac{\varepsilon^3}{6} \simeq \frac{4}{3}\tau^3. \quad \square$$

Solution 4 (normalization of the gauge-averaged measure).

$$\sum_J Z_{\beta_p}(J) = \sum_J \sum_{\sigma} e^{\beta_p \sum J \sigma \sigma} = \sum_{\sigma} \prod_{\langle ij \rangle} \sum_{J_{ij}=\pm 1} e^{\beta_p J_{ij} \sigma_i \sigma_j} = \sum_{\sigma} (2 \cosh \beta_p)^{N_B} = 2^N (2 \cosh \beta_p)^{N_B},$$

where the bond sum is performed after the substitution $J_{ij} \rightarrow J_{ij} \sigma_i \sigma_j$ (a bijection of the summation domain at fixed σ), and $\cosh$ is even. $\square$

Solution 5 (internal energy on the NL). $A(J) = \langle J_{ij} s_i s_j \rangle_{\beta_p, J}$ is gauge invariant, so it may be averaged with the gauge-averaged measure of Solution 4:

$$\mathbb{E}[\langle J_{ij} s_i s_j \rangle] = \sum_J \frac{Z_{\beta_p}(J) \langle J_{ij} s_i s_j \rangle_{\beta_p, J}}{2^N (2 \cosh \beta_p)^{N_B}} = \sum_J \frac{\partial Z_{\beta_p}(J) / \partial K_{ij}}{2^N (2 \cosh \beta_p)^{N_B}} \Big|_{K \equiv \beta_p},$$

since $Z \langle J_{ij} s_i s_j \rangle = \partial Z / \partial K_{ij}$ for a bond-resolved coupling K_{ij} . Exchanging $\sum_J$ and $\partial_{K_{ij}}$ and using, as in Solution 4, $\sum_J Z = 2^N \prod_b 2 \cosh K_b$, the derivative gives $2^N (2 \cosh \beta_p)^{N_B} \tanh \beta_p$. Hence $\mathbb{E}[\langle J_{ij} s_i s_j \rangle] = \tanh \beta_p$ and $U = -N_B \tanh \beta_p$: an analytic function of temperature along the line, including the multicritical point. $\square$

Solution 6 (the $\pi^2/12$ integral). $\ln(1 + e^{-u}) = \sum_{k \geq 1} (-1)^{k+1} e^{-ku} / k$, and term-by-term integration gives $\sum_{k \geq 1} (-1)^{k+1} / k^2 = \eta(2) = \pi^2/12$, with η the Dirichlet eta function. The remainder term is therefore $-\pi^2 T^2 / (12\sqrt{2\pi}J) + O(T^3)$; its T -derivative vanishes at $T = 0$, so it does not contribute to $S(0)$. $\square$

Solution 7 (near-critical order parameter). From Solution 3, $1 = \beta^2 - 2\beta^4 q + \frac{17}{3}\beta^6 q^2$, i.e. $q = \varepsilon / (2\beta^4) + \frac{17}{6}\beta^2 q^2$. Iterating with $\beta^4 = 1 + 2\varepsilon + O(\varepsilon^2)$: $q = \frac{\varepsilon}{2}(1 - 2\varepsilon) + \frac{17}{24}\varepsilon^2 = \frac{\varepsilon}{2} - \frac{7}{24}\varepsilon^2$. In terms of temperature, $\varepsilon = 2\tau + 3\tau^2 + O(\tau^3)$ (restoring J : $\beta J = 1/(1 - \tau)$), so

$$q = \tau + \frac{1}{3}\tau^2 + O(\tau^3),$$

and a nonzero solution first appears at $\varepsilon = 0$, i.e. $T_c = J$. $\square$

Solution 8 (1RSB overlap distribution). At integer n the 1RSB matrix has, per replica, $m_1 - 1$ intra-block partners and $n - m_1$ inter-block partners, so among the $n(n - 1)$ ordered pairs a fraction $(m_1 - 1)/(n - 1)$ carries q_1 and $(n - m_1)/(n - 1)$ carries q_0 . Continuing to $n \rightarrow 0$: $(m_1 - 1)/(0 - 1) = 1 - m_1$ and $(0 - m_1)/(0 - 1) = m_1$, giving $P(q) = m_1 \delta(q - q_0) + (1 - m_1) \delta(q - q_1)$. The two coefficients sum to 1 automatically; interpreting them as probabilities requires both to be nonnegative, i.e. $0 \leq m_1 \leq 1$, which is the range of the variational parameter in the replica limit, where it plays the role of the cumulative weight $x(q)$ of the Parisi function. $\square$

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